DevMatch AI: CRUD in VS Code PowerShell

Run every command below in the VS Code PowerShell terminal. The SQL is passed to the MySQL client by PowerShell, so you do not need MySQL Workbench or an interactive mysql> prompt.

Before you begin

Open the DevMatch project folder in VS Code. Use two PowerShell terminals: Terminal 1 runs the website; Terminal 2 sends SQL commands to MySQL. The local MySQL server must also be running.

Terminal 1 - start the website

```
cd D:\DBMS_DevApp\DevMatch_AI_DBMS_Mini_Project\devmatch-ai
& .\.venv\Scripts\python.exe -m flask --app app run --debug --port 5001
```

If port 5001 is already in use, the website is already running. Leave Terminal 1 open. In your browser, open **http://127.0.0.1:5001/dbms-demo**.

Terminal 2 - go to the project folder

```
cd D:\DBMS_DevApp\DevMatch_AI_DBMS_Mini_Project\devmatch-ai
```

If a SQL command returns no text, it usually succeeded. The SELECT commands below print the rows so you can confirm the result.

1. READ: confirm there are at least five existing jobs

```
$sql = "SELECT id, title, location FROM jobs ORDER BY id LIMIT 5;"
& .\.venv\Scripts\python.exe .\mysql_shell.py --execute $sql
```

Expected: five rows with IDs, job titles, and locations. The website already shows these records. On the DBMS demo page, click **Refresh table** and clear any old search filter.

2. CREATE: add five different demo jobs

In Terminal 2, paste this entire block. PowerShell will store the SQL text in **$sql** and run it through the project's MySQL client.

```
$sql = @"
INSERT INTO jobs
(recruiter_id, title, description, location, job_type)
VALUES
(1, 'PS Demo Python Developer', 'Build Python APIs',
    'Bengaluru', 'Full-time'),
(1, 'PS Demo Java Developer', 'Build Java services',
    'Chennai', 'Full-time'),
(1, 'PS Demo Data Analyst', 'Analyze business data',
    'Hyderabad', 'Internship'),
(1, 'PS Demo Web Designer', 'Design website pages',
    'Pune', 'Part-time'),
(1, 'PS Demo Cloud Engineer', 'Manage cloud systems',
    'Mumbai', 'Full-time');
"@
& .\.venv\Scripts\python.exe .\mysql_shell.py --execute $sql
```

On the website, enter **PS Demo** in the title search and click **Search**. You should see the five new rows. Click **Refresh table** after later commands.

3. READ: see the same five rows in PowerShell

```
$sql = "SELECT id, title, location, job_type FROM jobs " +
       "WHERE title LIKE 'PS Demo %' ORDER BY id DESC LIMIT 5;"
& .\.venv\Scripts\python.exe .\mysql_shell.py --execute $sql
```

Expected: the new five titles and their locations. Compare the IDs and values with the website table.

4. UPDATE: change one job

```
$sql = "UPDATE jobs SET title = 'PS Demo Senior Data Analyst', " +
       "location = 'Delhi' WHERE title = 'PS Demo Data Analyst' " +
       "ORDER BY id DESC LIMIT 1;"
& .\.venv\Scripts\python.exe .\mysql_shell.py --execute $sql
```

Check the updated row in PowerShell:

```
$sql = "SELECT id, title, location FROM jobs WHERE title = " +
       "'PS Demo Senior Data Analyst' ORDER BY id DESC LIMIT 1;"
& .\.venv\Scripts\python.exe .\mysql_shell.py --execute $sql
```

Expected: **PS Demo Senior Data Analyst** and **Delhi**. Refresh the website table; the same row now shows the changed title and location.

5. CREATE a temporary row, then DELETE it

```
$sql = "INSERT INTO jobs (recruiter_id, title, description, " +
       "location, job_type) VALUES (1, 'PS Demo Temporary Tester', " +
       "'Temporary record for delete demo', 'Remote', 'Full-time');"
& .\.venv\Scripts\python.exe .\mysql_shell.py --execute $sql
```

Refresh the website: the temporary tester appears, so six demo rows are visible. Now delete only that temporary row:

```
$sql = "DELETE FROM jobs WHERE title = 'PS Demo Temporary Tester' " +
       "ORDER BY id DESC LIMIT 1;"
& .\.venv\Scripts\python.exe .\mysql_shell.py --execute $sql
```

Check the result in PowerShell:

```
$sql = "SELECT COUNT(*) AS demo_rows FROM jobs " +
       "WHERE title LIKE 'PS Demo %';"
& .\.venv\Scripts\python.exe .\mysql_shell.py --execute $sql
```

Expected: **demo_rows = 5** if this is your first run. Refresh the website: the temporary tester disappears, and the five other demo jobs remain.

What should change on the website?

Operation	PowerShell result	Website after Refresh table
CREATE 5	SELECT lists 5 new rows	Five PS Demo rows appear
READ	SELECT shows saved values	Same IDs and values in table
UPDATE	Delhi shown for Data Analyst	Data Analyst row changes to Delhi
CREATE temp	Temporary row inserted	Temporary tester appears
DELETE temp	Count returns to 5	Temporary tester disappears

If the website does not change

1. Make sure you opened **/dbms-demo** on port **5001**. 2. Click **Refresh table**. 3. Clear any search other than **PS Demo**. 4. Run the READ command to confirm MySQL contains the row. 5. Check that the website process is still running in Terminal 1.

If MySQL is not running

From a PowerShell terminal, start the portable local MySQL server, then repeat the SQL command:

```
Start-Process -FilePath "D:\DBMS_DevApp\mysql-local\mysql-8.4.11-winx64\bin\mysqld.exe" `
  -ArgumentList @(
    "--basedir=D:\DBMS_DevApp\mysql-local\mysql-8.4.11-winx64",
    "--datadir=D:\DBMS_DevApp\mysql-local\data",
    "--bind-address=127.0.0.1", "--port=3307") `
  -WindowStyle Hidden
```

Running the CREATE steps again will add another set of demo jobs. The UPDATE and DELETE steps target the latest matching row. To exit the website server, press **Ctrl+C** in Terminal 1.